

Measuring Scalar Constructs in Social Science with LLMs

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Abstract

Many constructs that characterize language, like its complexity or emotionality, have a naturally continuous semantic structure; a public speech is not just “simple” or “complex,” but exists on a continuum between extremes. Although large language models (LLMs) are an attractive tool for measuring scalar constructs, their idiosyncratic treatment of numerical outputs raises questions of how to best apply them. We address these questions with a comprehensive evaluation of LLM-based approaches to scalar construct measurement in social science. Using multiple datasets sourced from the political science literature, we evaluate four approaches: unweighted direct pointwise scoring, aggregation of pairwise comparisons, token-probability-weighted pointwise scoring, and finetuning. Our study yields actionable findings for applied researchers. First, LLMs prompted to generate pointwise scores directly from texts produce discontinuous distributions with bunching at arbitrary numbers. The quality of the measurements improves with pairwise comparisons made by LLMs, but it improves even more by taking pointwise scores and weighting them by token probability. Finally, finetuning smaller models with as few as 1,000 training pairs can match or exceed the performance of prompted LLMs.

1 Introduction

While many constructs in the social sciences are treated as categorical, such as the TOPIC of a speech, others are more appropriately considered as a continuum, like the EMOTIONAL INTENSITY of that speech (e.g. Gennaro and Ash, 2022). Valid scalar measurement of such constructs enables a wide range of substantive applications in social science research, such as modeling legislator behavior (Poole and Rosenthal, 2001) or analyzing polarization in immigration debates (Card et al.,

*Equal contribution

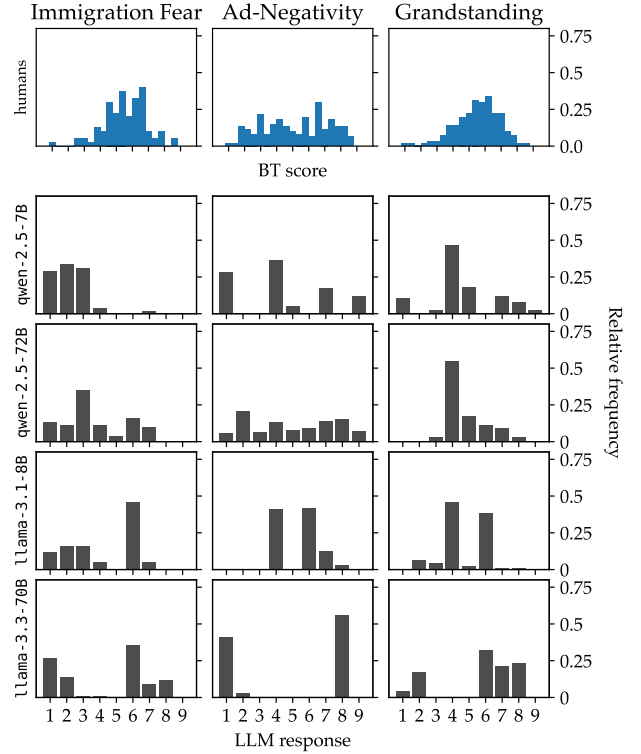


Figure 1: Distributions of LLM scores for scalar constructs do not align with the reference distribution, nor do they correspond between models. *Top*: Distribution of text items’ scores on latent dimension for three different tasks estimated by fitting a Bradley-Terry (BT) model to human-annotated pairwise comparisons between text items. *Bottom*: Distribution of the scores different LLMs’ assign to the same text items if prompted to score them on a 1–9 scale.

2022). Assigning scalar values to texts is therefore a fundamental task in computational text analysis (Grimmer and Stewart, 2013).

The standard array of NLP methods have been brought to bear on this measurement problem: supervised methods using bag-of-words representations (Laver et al., 2003; Gentzkow et al., 2019); unsupervised models that assume a latent variable corresponding to the construct (Monroe and Maeda,

2004; Slapin and Proksch, 2008; Vafa et al., 2020; Hofmann et al., 2022; Stoehr et al., 2023); and large language models (LLMs; e.g. Röttger et al., 2024; Le Mens and Gallego, 2025; Kim et al., 2025).

LLMs in particular are an attractive solution for assigning scalar measurements to texts, because in-context learning (Brown et al., 2020) requires little or no task-specific training data. However, the space of possible approaches to scoring texts with LLMs is large, and naive prompting can lead to unreliable results (Wang et al., 2024; Röttger et al., 2024). Take the zero-shot setting, where an LLM is instructed to score texts on an ordinal scale (e.g., 1–9). Models tend to produce “heaped” distributions for this prediction task, wherein probability mass is concentrated only on a few numeric tokens (fig. 1). This behavior is likely due to systematic biases favoring certain tokens induced during pre- or post-training (Zhao et al., 2021; Razeghi et al., 2022).¹ This behavior may mislead researchers studying the absolute level of, or distance between, observations of that construct, pointing to the need to explore alternative approaches.

One such alternative is the *pairwise ranking* of items, where the abstract construct is operationalized as a per-item latent variable that generates observed ranks. As in the case of human annotation, an LLM compares pairs of texts in terms of their intensity on an underlying scale (Wu et al., 2023; Stoehr et al., 2024). The latent per-item scores are then estimated with probabilistic models of ranked pairs, like Bradley-Terry (Bradley and Terry, 1952).

With human coders, pairwise comparisons produce text rankings that are more robust than annotators’ direct ratings of individual text items (Kendall 1948; Kingsley and Brown 2010; De Bruyne et al. 2021; Narimanzadeh et al. 2023; cf. Wood et al. 2018). Accordingly, pairwise comparison has been applied to measure various constructs in social science research (Benoit et al., 2019; Carlson and Montgomery, 2017) or to validate such measures (Gennaro and Ash, 2022; Hargrave and Blumenau, 2022). In NLP, pairwise (or listwise) comparisons are common in human annotation (Lopez, 2012; Sakaguchi et al., 2014; Sakaguchi and Van Durme, 2018; Simpson and Gurevych, 2018; Chen et al., 2021; De Bruyne et al., 2021; Narimanzadeh et al.,

2023; Qin et al., 2023; Stoehr et al., 2024), automated system evaluation (Liusie et al., 2024; Zheng et al., 2023), and preference modeling (Ziegler et al., 2019; Ouyang et al., 2022).

Despite the appeal of pairwise comparisons for measuring social science constructs, we lack comparative evidence of its utility in automated scalar measurement with LLMs. We therefore present a comprehensive evaluation of LLM prompting and finetuning methods for text scoring. Using three human-labeled datasets from two political science studies (Carlson and Montgomery, 2017; Park, 2021) that cover different target constructs, we consider various prompting methods and calibration techniques—combinations of direct scoring and pairwise comparisons, drawing from the literature on LLM evaluators (Wang et al., 2025). In addition to prompting, we adapt reward modeling methods (Ziegler et al., 2019; Ouyang et al., 2022) to finetune models on pairwise data, comparing them to standard regression finetuning.

We find that the benefit of pairwise comparisons depends on whether the models are prompted or finetuned. For in-context learning, direct pointwise scoring of text items can be just as (or more) effective than pairwise comparison (table 2), but only after computing a probability-weighted average over the ordinal tokens (see Wang et al., 2025). However, fine-tuning a reward model with as few as 1,000 labeled pairs can produce scoring models that outperform a prompted model (table 3), even when the prompted model has two orders of magnitude more parameters (finetuned *regression* models, on the other hand, require more data).

Summarizing our contributions, we:

- Analyze issues with direct pointwise scoring and LLMs’ “heaped” responses (§2.1).
- Compile a text scoring benchmark of datasets from the social science literature (§3.1).
- Evaluate a suite of LLMs in zero-, few-shot, and fine-tuning settings, comparing pairwise and pointwise scoring (§4).
- Provide actionable recommendations for practitioners (§4).

2 Scoring Text Items

We compare common approaches to scoring text items with LLMs. First, we cover *pointwise scoring* via in-context learning (ICL), where the model scores *individual texts* in isolation, and discuss shortcomings of this setup. We then turn to *pair-*

¹Round numbers are far more common: in the Dolma (Soldaini et al., 2024) pretraining set, the n-gram 25 percent appears roughly 5M times, compared to about 1M for 24 or 26 percent, per the WIMBD tool from Elazar et al. (2024)

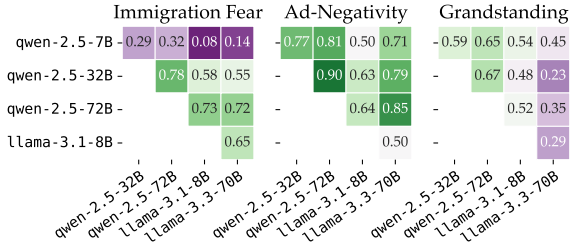


Figure 2: Inter-model agreement (Krippendorff’s α) by dataset for zero-shot pointwise prompting. Even within model families, agreement can be relatively low.

wise comparisons with ICL, where the model compares *pairs of texts*. Last, we discuss finetuning procedures to apply when training data is available.

2.1 Pointwise prompting with LLMs

Generative LLMs define conditional probability distributions $P(y \mid \mathbf{x})$, where $\mathbf{x} \in \Sigma^*$ is a prompt and $y \in \Sigma$ is the next token from the vocabulary Σ to be generated. The most straightforward way to score a text item, therefore, is to prompt an LLM to output a value on a fixed, pre-defined scale, such as 1–9 (Tian et al., 2023; O’Hagan and Schein, 2024; Ziems et al., 2024; Le Mens and Gallego, 2025). The researcher defines the scale in terms of discrete answer token candidates $\mathcal{C} \subset \Sigma$ (e.g., $\mathcal{C} = \{1, \dots, 9\}$) and prompts the LLM to assign each text one of the available scale point values. This *pointwise* prompting strategy can be further refined by including detailed coding instructions in the prompt (Ruckdeschel, 2025), or by averaging scores over multiple sentences within a document (Le Mens and Gallego, 2025).

More recently, Wang et al. (2025) show that instead of relying on an LLM’s response directly, the weighted average of a model’s token probabilities produces more accurate scores in an LLM-as-judge setting (following up on findings from Liu et al. 2023; Yasunaga et al. 2024; Lee et al. 2024). Specifically, each answer token candidate in the numerical scale is multiplied by its generation probability, then summed:

$$\sum_{c \in \mathcal{C}} P(y = c \mid \mathbf{x}) \cdot c, \quad (1)$$

where P is the probability the LLM assigns to token $y = c$ given prompt $\mathbf{x}$.

Pitfalls of pointwise prompting. Prompting LLMs to directly score individual text items has several limitations (O’Hagan and Schein, 2023).

First, the scores that generative LLMs assign to individual text items tend to be poorly calibrated: common token bias (Zhang et al., 2024) and prompt phrasing (Sclar et al., 2024) can dramatically affect models’ responses. See Figure 1: for the three datasets we study (section 3.1), the zero-shot pointwise scoring outputs of different LLMs produce distributions over items that do not agree with scores inferred from ground-truth human annotations using Bradley-Terry (see section 2.2).

Consider the results for the IMMIGRATION FEAR data, which focuses on survey respondents’ anxieties about immigration in the U.S. (Carlson and Montgomery, 2017, see Section 3.1). The reference distribution is centered around the midpoint of the inferred scale, bimodal, and symmetric. None of the distributions of LLM scores align with this reference. The responses of Qwen 2.5 models (Qwen Team, 2025), for example, tend to be right-skewed, especially for the smaller 7B variant. And while both Llama 3 models’ (Meta, 2024) responses are bimodal, they do not match the symmetry in the reference distribution.

Figure 1 also illustrates a second pathology. When LLMs are asked to score texts, their responses can exhibit a phenomenon known as *heaping*, where model outputs are concentrated on particular values, rather than using the full extent of the scale (see Roberts and Brewer, 2001).² For example, Llama-3.3-70B outputs scores $y = 1$ and $y = 8$ for most of the text items in the AD-NEGATIVITY data.

Of course, not all these response distributions can be true at the same time, undermining the reliability of LLMs’ zero-shot pointwise scoring responses. Accordingly, models’ responses often agree at best moderately, even within a model family (see fig. 2).

Calibration could potentially be improved by taking the probability-weighted average over numerical tokens in the scale (eq. 1), as it could smooth the distributions, rendering them more like the reference. However, LLMs’ confidence over tokens is often poorly calibrated and heavily concentrated on the modal response (Tian et al., 2023; Xie et al., 2024), which can distort the resulting weighted average (fig. 3). In addition, confidences are strongly

²Heaping is especially likely when using fine-grained scales, such as 0–100, because models often choose scores divisible by 10 or 5 (e.g., Le Mens and Gallego, 2025). For LLMs, heaping can be caused by a lack of explicit diversity incentives during training or bias in the training data (Zhang et al., 2024).

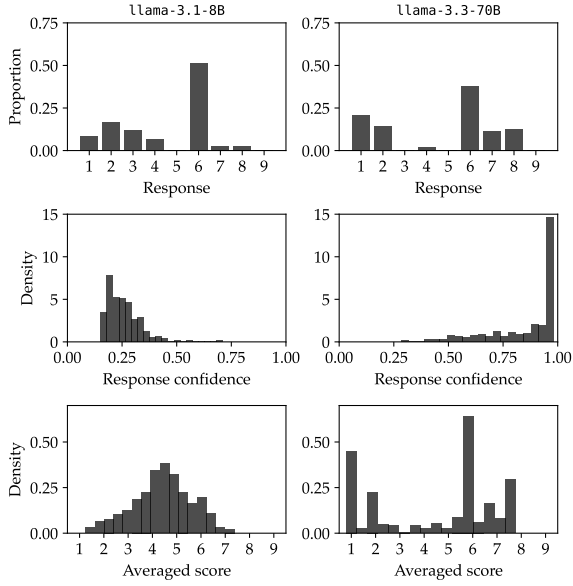


Figure 3: Distribution of Llama-3.1-8B and Llama-3.3-70B’s modal responses (top row), confidences in modal responses (mid row), and probability-weighted average responses bottom row) from zero-shot pointwise prompting for texts in the IMMIGRATION FEAR data (see §3.1). While both LLMs modal response distributions exhibit similar levels of concentration on $y = 6$, the larger model variant (right) tends to be much more confident in its response, which reduces the smoothing effect probability-weighted averaging has on its response distribution.

affected by the model size and prompting method (fig. 6 in the Appendix).

Although prior work suggests strong correlations between pointwise LLM scores and human ground truth at the document level for certain constructs (Le Mens and Gallego, 2025), the above results indicate that LLMs are not a panacea. In the remainder of this work, we systematically assess the reliability of scoring text via pointwise prompting, pairwise prompting, and finetuning.

2.2 Pairwise Comparisons

An alternative to pointwise scoring is collecting pairwise preferences. Given a pair of text items, a human annotator or LLM selects the item that better exemplifies the target construct or that is more extreme on the underlying dimension. Given multiple such comparisons, a probabilistic pairwise ranking model such as Bradley-Terry (Bradley and Terry, 1952, henceforth referred to as BT) can be used to estimate items’ location on the latent construct. The BT model takes the pairwise outcomes as input and estimates the latent “strength” of each

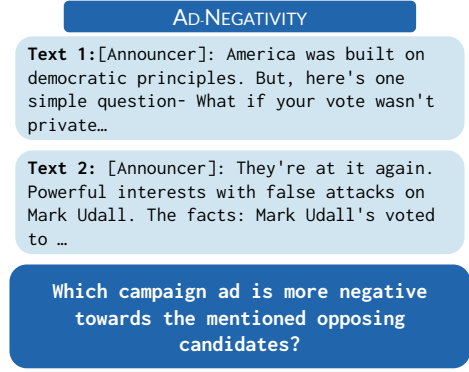


Figure 4: Pairwise comparison places two text items relative to one another regarding a given construct.

text item via maximum likelihood, modeling the win probability with a logistic link function:³

$$P(i > j) = \frac{e^{z_i}}{e^{z_i} + e^{z_j}}. \quad (2)$$

for texts $\mathbf{x}_i, \mathbf{x}_j$ and their corresponding latent scores z_i, z_j .

For different social science constructs measured with human annotations, scores inferred from annotators’ pairwise comparisons measure the target dimensions more reliably than ratings obtained from expert annotations through direct, pointwise annotation (Carlson and Montgomery, 2017). More recently, pairwise judgments obtained from LLMs have also been leveraged to estimate constructs in social science applications (Sarkar et al., 2025; Wu et al., 2024).⁴ However, a gap remains in the literature: studies using pairwise comparisons often lack adequate comparisons to pointwise scoring baselines (Sarkar et al., 2025), and vice versa (Le Mens and Gallego, 2025; O’Hagan and Schein, 2024).

2.3 Finetuning

If labeled data are available, another option is *finetuning* LLMs (Howard and Ruder, 2018). When the labeled data are annotated pairs (as is the case here, see §3.1), we adapt *reward modeling* objectives, used to align language models to human preferences with pairwise data (Christiano et al., 2017; Ziegler et al., 2019; Ouyang et al., 2022). Per Ouyang et al., the loss is the negative-log likelihood

³There are many probabilistic rank models (Malloes, 1957; Luce, 1959; Elo, 1967; Plackett, 1975; Herbrich et al., 2006; Lu and Boutilier, 2011; Carlson and Montgomery, 2017); we focus on Bradley-Terry due to its simplicity and ubiquity.

⁴There is a computational downside: direct scoring is $\mathcal{O}(n)$, whereas pairwise is $\mathcal{O}(n^2)$, although subsampling can improve this to $\mathcal{O}(kn)$.

of a pairwise comparison under Bradley-Terry,

$$\ell_{\theta}(\mathbf{x}_h, \mathbf{x}_l) = -\log(\sigma(r_{\theta}(\mathbf{x}_h) - r_{\theta}(\mathbf{x}_l))), \quad (3)$$

with $\mathbf{x}_h$ being the preferred item in the pair (over $\mathbf{x}_l$), and $r_{\theta}(\cdot)$ being an LLM with a regression head. The score for an item $\mathbf{x}_i$ is then $r_{\theta}(i)$, and the model is trained via gradient descent. An alternative is regression on available scores directly, for example with a mean squared error loss.

3 Experimental Setup

Given the established benefits of using pairwise annotations in social science data, our datasets all consist of pairwise comparison data from the social science literature. First, we outline each dataset (statistics in table 1), then discuss methods, LLM variants, evaluation strategy, and metrics.

3.1 Datasets

IMMIGRATION FEAR Carlson and Montgomery (2017) rely on trained crowdworkers to measure *the level of fear, anxiety, or worry toward immigration or immigrants in the U.S.* expressed in responses to an open-ended survey question. The construct targeted in this dataset relates to other research on the use of (discrete) emotions in political communication (Widmann and Wich, 2022; Gennaro and Ash, 2022).

AD-NEGATIVITY In the same work, Carlson and Montgomery also crowd-source pairwise comparisons to measure *the level of negativity of political campaign advertisements*, a construct related to negative campaigning and other attack behaviors (Walter and Nai, 2015; Licht et al., 2025). The ads in their data were aired before the 2008 U.S. Senate elections and obtained from the Wisconsin Advertising Project (WiscAds) database.⁵ The annotations come from trained online workers, who indicate which ad in a pair is “most negative towards” or “least positive about” the “candidate(s) mentioned” (Carlson and Montgomery, 2017, p. 828).⁶

GRANDSTANDING Last, we use a dataset compiled by Park (2021), who measures speakers’ *grandstanding* in House committee hearings in the U.S. Congress. Park defines grandstanding as opinionized speech behavior that “sends political mes-

⁵elections.wisc.edu/wisconsin-advertising-project/

⁶Ornstein et al. (2025) use experts’ ad-level (pointwise) ratings to evaluate LLMs’ pointwise scoring in this data.

	Items	Pairs	Node degree statistics		
			Co.	Dens.	μ (σ)
IMMIGRATION FEAR	334	6,489	33	0.11	37.6 (1.5)
AD-NEGATIVITY	935	9,489	18	0.02	20.2 (0.9)
GRANDSTANDING	3,499	38,348	17	0.01	21.8 (7.6)

Table 1: Datasets consist of annotated pairs of items, forming a graph. Connectivity (Co.) is the number vertices (*items*) needed to disconnect the graph; density (Dens.) is the ratio of observed edges (*pairs*) to possible edges. μ (σ) are the mean (std. dev.) degree per node.

sages by taking positions on policy issues or framing the image of a party or the administration” and contrasts it with information-seeking, fact-oriented speech behavior. The texts are short statements sampled from roughly 12,000 speeches held by House committee members during public hearings in the 105th to 114th Congresses. Trained online workers indicate which statement in a pair would be better described as opinionized or grandstanding as opposed to fact-based or information seeking.

3.2 Methods and Models

Prompting. Prompted models infer either pointwise scores per item or pairwise ranks per pair. For pointwise prompting, we instruct the model to provide an integer on a 1–9 scale per construct.⁷ To produce a final score, we apply the probability weighting from eq. (1).

In the pairwise case, the model selects the item that is more extreme for the construct. We borrow from Wang et al. (2025) and prompt the model with different orderings to avoid positional biases (Zhao et al., 2021; Han et al., 2023; Wang et al., 2023). We record the distribution over the choice tokens and take the average (see Appendix A.2). In both settings, we run zero-shot and random 5-shot exemplars.⁸ The instructions are derived from the original publications’ codebooks (prompts in appendix D).

We use instruction-tuned open-weight LLMs in different sizes: Llama 3.1 8B, Llama 3.3 70B (Meta, 2024) as well as the 7B, 32B, and 72B variants of Qwen 2.5 (Qwen Team, 2025). All models were run with 4-bit quantization.⁹

⁷We use a range between 1 and 9 because (some) models tokenize two-digit numbers into two tokens.

⁸For pointwise scores, using anchors for the different scale points did not consistently improve metrics over 5-shot.

⁹Using the transformers (Wolf et al., 2020) and bitsandbytes (see Dettmers et al., 2023). Our results are robust to using no quantization (Table 9). GPU specifications are reported in Table 6.

Finetuning. Using the reward modeling objective in eq. (3), we finetune DeBERTa-v3-large (He et al., 2021), ModernBERT-large (Warner et al., 2024), and Llama-3.1-8B-Instruct models. The first two models are encoder-only models commonly used to fine-tune classifiers in political text analysis (cf. Timoneda and Vallejo Vera, 2024); Llama is a decoder-only generative language model with double the parameters of DeBERTa-v3-large. To finetune the 8B Llama model, we use QLoRA with 4-bit quantization (Detrmers et al., 2023). We sweep over the learning rate for each model to maximize pairwise accuracy on a validation set (details in appendix B.2).

Further, to compare pairwise to pointwise finetuning, we finetune DeBERTa-v3-large with a regression head. The labels are scores estimated from BT models fit to the human pairwise annotations in the training set.

3.3 Evaluation Strategy

Each dataset contains a unique set of text *items*; labels between pairs indicate which item is more extreme on the target scalar construct (e.g., which of two campaign ad transcripts is more negative about an opponent). Therefore, the pairwise labeled data in each dataset form a directed connected graph $G = (V, E)$ with vertices representing text items and edges representing comparison relations.¹⁰

The graphs are connected, so constructing a train-test split and an adequate evaluation strategy is a somewhat delicate operation. We proceed as follows. First sample $n_{\text{test}} = 100$ high-degree held-out evaluation vertices $V_{\text{test}} \in V$, with train vertices $V_{\text{train}} := V \setminus V_{\text{test}}$. The induced subgraph $G_{\text{train}} := G[V_{\text{train}}]$ comprises the edges and items used for training the finetuned models. $G_{\text{eval}} := G[E \setminus E_{\text{train}}]$ is then the graph of all edges that contain at least one test vertex V_{test} (i.e., the graph also contains items from V_{train}).

Evaluation Metrics. Our primary focus is on LLMs’ *scoring performance*. To estimate items’ ground-truth reference scores, we fit a Bradley-Terry (BT) model¹¹ to the entire graph G , resulting in item-level scores $r_{\text{BT}}(x_i)$ for all $x_i \in V$. We can then evaluate how well an LLM can predict these

item-level scores. Specifically, we measure scoring performance with **Spearman rank correlation** ρ and the root mean squared error (**RMSE**) in the subset of items in V_{test} .¹² ρ measures how well an LLM ranks text items relative to the reference scores. RMSE measures the average magnitude of the errors between text items’ predicted and reference scores. Both metrics complement each other: ρ focuses on ranking consistency, while RMSE evaluates the precision of the actual numerical predictions.

A secondary focus is on *pair classification performance*, i.e., whether models can predict the pairwise labels $\mathbb{I}[x_i > x_j]$, $(x_i, x_j) \in E$. We measure classification performance with **accuracy**, which we compute against *all* edges in G_{eval} —that is, any edge where a test vertex appears.¹³

Influence of training data size. For the finetuning experiments, we also evaluate the effect of increasing the number of edges, while controlling for differences in dataset structure (table 1). Our algorithm iteratively adds edges to each of the three graphs such that the average degree and clustering coefficient remain the same for each n , up until about 2000 edges, where maintaining graph similarity is no longer feasible given the differing structural characteristics of our datasets.

4 Results

First, we report results from prompting and then fine-tuning. Overall, pairwise comparison does not improve prompting results (see table 2), but it does help in finetuning (fig. 10).

4.1 Prompting

Table 2 compares pairwise and pointwise prompting approaches for text scoring. We report results for both zero- and 5-shot prompting with probability-weighted averaging.¹⁴ We summarize our main insights below.

¹²This avoids leakage when evaluating finetuned models and makes evaluations of prompted and finetuned models comparable.

¹³This choice biases the finetuned accuracies upward because they have seen the train vertices, but there are relatively few edges with both $u, v \in V_{\text{test}}$.

¹⁴Using probability-weighted averages instead of models’ modal response tends to improve the accuracy and scoring performance for all models and few-shot settings in the GRANDSTANDING data and the AD-NEGATIVITY data (but one) and for most models and most few-shot settings in the IMMIGRATION FEAR data for accuracy and RMSE.

¹⁰This structure is unlike standard LLM paired preference datasets, which are disconnected. There, comparisons between generated texts are conditioned on some shared context, like an instruction, making them incomparable across contexts.

¹¹Using the `choix` implementation (Maystre et al., 2022)

		IMMIGRATION FEAR			AD-NEGATIVITY			GRANDSTANDING		
	Shots	Acc	ρ	RMSE	Acc	ρ	RMSE	Acc	ρ	RMSE
Qwen-2.5-7B										
pairwise	0-shot	0.65±0.01	0.53±0.07	0.18±0.01	0.76±0.01	0.83±0.02	0.20±0.01	0.62±0.01	0.45±0.04	0.21±0.01
	5-shot	0.73±0.01	0.81±0.03	0.15±0.01	0.77±0.01	0.84±0.02	0.18±0.01	0.61±0.01	0.40±0.05	0.23±0.01
pointwise	0-shot	0.70±0.01	0.63±0.06	0.40±0.01	0.78±0.01	0.85±0.02	0.19±0.01	0.65±0.01	0.58±0.05	0.22±0.01
	5-shot	0.75±0.01	0.81±0.04	0.26±0.01	0.79±0.01	0.87±0.02	0.20±0.01	0.65±0.01	0.57±0.05	0.24±0.01
Qwen-2.5-32B										
pairwise	0-shot	0.71±0.01	0.68±0.04	0.26±0.01	0.80±0.01	0.85±0.02	0.18±0.01	0.66±0.01	0.51±0.04	0.20±0.01
	5-shot	0.77±0.01	<u>0.85±0.03</u>	0.16±0.01	<u>0.80±0.01</u>	0.85±0.02	0.18±0.01	<u>0.68±0.01</u>	0.59±0.03	0.19±0.01
pointwise	0-shot	0.72±0.01	0.71±0.04	0.26±0.01	0.79±0.01	0.87±0.02	0.14±0.01	0.65±0.01	0.58±0.04	0.31±0.01
	5-shot	<u>0.77±0.01</u>	<u>0.85±0.03</u>	0.22±0.01	<u>0.80±0.01</u>	0.90±0.02	0.16±0.01	0.66±0.01	0.61±0.05	0.30±0.01
Qwen-2.5-72B										
pairwise	0-shot	0.73±0.01	0.72±0.05	0.19±0.01	<u>0.80±0.01</u>	0.86±0.02	0.18±0.01	0.66±0.01	0.48±0.04	0.20±0.01
	5-shot	0.77±0.01	0.84±0.03	0.16±0.01	<u>0.81±0.01</u>	0.87±0.02	0.19±0.01	0.67±0.01	0.56±0.03	0.19±0.01
pointwise	0-shot	0.76±0.01	0.81±0.03	0.23±0.01	<u>0.81±0.01</u>	0.91±0.01	0.13±0.01	0.65±0.01	0.60±0.04	0.21±0.01
	5-shot	<u>0.77±0.01</u>	0.87±0.03	0.20±0.01	0.81±0.01	0.92±0.01	0.17±0.01	0.67±0.01	0.66±0.04	0.29±0.01
Llama-3.1-8B										
pairwise	0-shot	0.58±0.01	0.48±0.09	0.32±0.01	0.59±0.01	0.46±0.08	0.26±0.02	0.55±0.01	0.21±0.07	0.25±0.01
	5-shot	0.73±0.01	0.75±0.05	0.21±0.01	0.74±0.01	0.74±0.04	0.21±0.01	0.59±0.01	0.36±0.06	0.23±0.01
pointwise	0-shot	0.71±0.01	0.68±0.06	0.18±0.01	0.77±0.01	0.82±0.02	0.24±0.01	0.64±0.01	0.57±0.04	0.16±0.01
	5-shot	0.74±0.01	0.78±0.04	0.27±0.01	<u>0.81±0.01</u>	0.90±0.01	0.20±0.01	0.66±0.01	0.61±0.05	0.39±0.01
Llama-3.3-70B										
pairwise	0-shot	0.73±0.01	0.74±0.05	0.18±0.01	0.78±0.01	0.83±0.03	0.19±0.01	0.66±0.01	0.54±0.03	0.19±0.01
	5-shot	0.76±0.01	<u>0.85±0.03</u>	<u>0.15±0.01</u>	0.79±0.01	0.84±0.02	0.19±0.01	0.67±0.01	0.54±0.03	0.20±0.01
pointwise	0-shot	0.73±0.01	0.75±0.04	0.30±0.01	0.80±0.01	0.89±0.01	0.29±0.01	0.66±0.01	0.55±0.04	0.26±0.01
	5-shot	0.76±0.01	0.84±0.03	0.28±0.01	<u>0.80±0.01</u>	0.90±0.02	0.25±0.01	0.69±0.01	<u>0.65±0.04</u>	0.30±0.01

Table 2: Comparison of pointwise scoring and pairwise comparison LLM prompting methods for scalar construct measurement. Item-level scores in pairwise comparison setup are inferred with Bradley-Terry (BT); pointwise scores use token probability-weighted averaging. Pointwise scoring tend to work better and is computationally cheaper. *Notes:* Results reported for 0-shot prompting (no exemplars) and 5-shot prompting (using five randomly sampled exemplars per dataset). Accuracy (Acc) measured relative to human ground-truth pairwise comparisons; Spearman’s rank correlation (ρ) and root mean squared error (RMSE) are relative to the ground-truth BT scores inferred from those comparisons. Values report averages $\pm$ one standard deviation computed based on 25 bootstrapped estimates in test split. Values in bold mark the best result for a dataset and metric and values underlined flag results within one standard error of the best result.

Pointwise outperforms pairwise prompting.

Across all three datasets, 5-shot pointwise prompting with probability-weighted averaging consistently outperforms or is equally reliable as zero- or 5-shot pairwise prompting (table 2).¹⁵ This denotes a key difference in human and LLM construct measurement. Humans demonstrate increased accuracy with pairwise comparisons over pointwise scoring due to reduced calibration errors (Carlson and Montgomery, 2017). This advantage does not translate to LLMs in our datasets, however. Unsurprisingly, it is the larger model variants that achieve these top scores (Qwen-2.5-72B for IMMIGRATION FEAR and AD-NEGATIVITY, and

Llama-3.3-70B for GRANDSTANDING). In line with previous research (Chamieh et al., 2024), 5-shot prompting substantially improves the pointwise scoring performance in terms of accuracy and correlation. The magnitude of improvement depends on model size (smaller models benefit more) and on the task (few-shot exemplars help more in GRANDSTANDING dataset).¹⁶

Correlation values can mask important differences in error magnitude. While correlation can be an effective measure of how well a model ranks text items relative to ground truth, looking at correlation alone fails to capture how predicted scores diverge from the true distribution. In the

¹⁵Using a proprietary model (GPT-4o) yields comparable results (see table 8).

¹⁶We experiment with other exemplar selection strategies, but did not see an improvement over choosing five at random.

	IMMIGRATION FEAR (<i>all</i> = 2729)			AD-NEGATIVITY (<i>all</i> = 6760)			GRANDSTANDING (<i>all</i> = 32308)		
N_{train}	Acc	ρ	RMSE	Acc	ρ	RMSE	Acc	ρ	RMSE
DeBERTa-v3-large									
500	0.74±0.02	0.79±0.10	0.28±0.02	0.78±0.01	0.86±0.02	0.23±0.02	0.64±0.01	0.67±0.04	0.22±0.04
1000	0.75±0.01	0.83±0.06	0.23±0.05	<u>0.80±0.01</u>	0.88±0.01	0.20±0.05	0.65±0.01	0.70±0.05	0.21±0.05
2000	0.77±0.01	<u>0.89±0.03</u>	<u>0.19±0.01</u>	<u>0.80±0.01</u>	0.89±0.02	0.19±0.05	0.66±0.01	0.73±0.03	0.22±0.09
<i>all</i>	0.78±0.00	0.91±0.02	<u>0.18±0.04</u>	0.81±0.01	0.91±0.00	0.19±0.03	0.68±0.01	0.78±0.03	0.16±0.02
ModernBERT-large									
500	0.74±0.01	0.79±0.04	<u>0.20±0.05</u>	0.74±0.02	0.77±0.04	0.20±0.04	0.61±0.03	0.51±0.07	0.22±0.03
1000	0.76±0.01	0.83±0.02	<u>0.20±0.07</u>	0.75±0.01	0.80±0.03	0.19±0.05	0.62±0.01	0.56±0.03	0.19±0.03
2000	0.77±0.01	0.85±0.03	0.17±0.04	0.78±0.02	0.83±0.04	<u>0.17±0.04</u>	0.64±0.01	0.62±0.05	0.19±0.05
<i>all</i>	0.77±0.01	0.85±0.01	<u>0.18±0.04</u>	0.79±0.02	0.87±0.03	<u>0.17±0.04</u>	0.66±0.01	0.72±0.03	0.21±0.05
Llama-3.1-8B-Instruct									
500	0.76±0.01	0.86±0.04	<u>0.20±0.08</u>	0.78±0.01	0.85±0.02	0.16±0.02	0.64±0.01	0.63±0.06	0.20±0.03
1000	0.77±0.01	0.86±0.03	<u>0.19±0.04</u>	<u>0.80±0.01</u>	0.88±0.01	<u>0.18±0.06</u>	0.63±0.02	0.60±0.09	0.19±0.03
2000	0.76±0.01	0.82±0.03	0.23±0.02	0.79±0.01	0.85±0.05	0.16±0.03	0.62±0.00	0.60±0.04	0.19±0.03
<i>all</i>	0.76±0.01	0.83±0.05	0.17±0.04	0.79±0.01	0.87±0.02	<u>0.17±0.05</u>	0.65±0.01	0.70±0.05	0.21±0.04
<i>best pointwise prompting results (see Table 2)</i>									
	0.77±0.01	0.87±0.03	0.18±0.01	0.81±0.01	0.92±0.01	0.13±0.01	0.69±0.01	0.66±0.04	0.16±0.01

Table 3: Reward model finetuning results by model, number of training examples, and dataset. Finetuning tends to outperform prompting after roughly 2,000 examples. *Notes:* Metrics reported are the pair classification accuracy (Acc), as well as the Spearman’s rank correlation (ρ) and the root mean squared error (RMSE) against the ground-truth BT scores. Values report are averages $\pm$ one standard deviation computed by summarizing results across five folds.

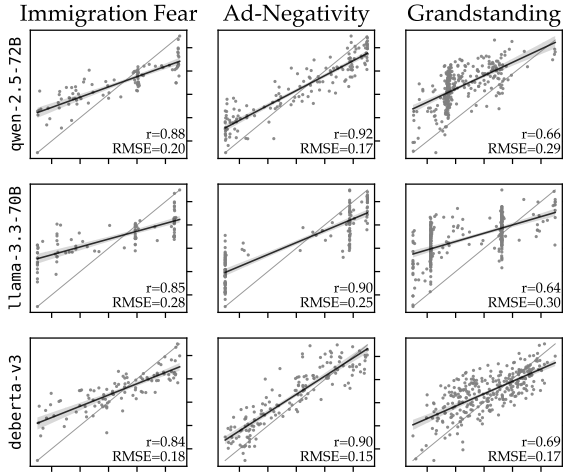


Figure 5: Relation between model responses and ground-truth BT scores. Qwen-2.5-72B and Llama-3.3-70B are prompted, showing probability-weighted average pointwise scores (5-shot). DeBERTa-v3 (large) has been finetuned on 2,000 training pairs for each datasets. These results show RMSE and Spearman’s ρ may not always agree.

AD-NEGATIVITY dataset, both Qwen-2.5-72B and Llama-3.3-70B (with 5-shot prompting) have high correlations. However, their RMSE scores reveal that Llama-3.3-70B makes larger

errors, while achieving a rank correlation similar to that of the other two models (Figure 5). Llama-3.3-70B’s worse RMSE scores can be attributed to the fact that, due to its extremely high response confidence (fig. 6), the heaping in its modal responses (fig. 1) is virtually unaffected by token probability weighting (fig. 3). Consequently, its scores are bunched around 0.0, 0.9, and 1.0.

This example underscores that, for scoring tasks, considering both RMSE and correlation provides a balanced view of model behavior. Specifically, the choice of a model should depend on a researcher’s downstream analysis and research design. If their analysis only requires that texts are sorted in the correct order, the strong tendency to produce heaped outputs of Llama-3.3-70B should be less of a concern. However, if their research design requires measuring the relative distances between observations, RMSE becomes more important, and Qwen-2.5-72B might be a better option.

4.2 Finetuning

Prompting billion-parameter LLMs requires substantial compute. Finetuning smaller LLMs can be a more efficient alternative for scoring text items

under different data constraints (table 3).

If labeled data is available, finetuning offers a compelling alternative to prompting. In IMMIGRATION FEAR, finetuning on as little as 500 pairs (with Llama-3.1-8B-Instruct) can yield almost as good correlation and equal RMSE as the most performant prompting approach (5-shot pointwise prompting with Qwen-2.5-72B). Overall, DeBERTa-v3-large finetuned on 2000 pairwise examples is comparable or better than almost all pointwise prompting approaches on accuracy and correlation. In the GRANDSTANDING dataset, finetuning DeBERTa-v3-large with just 500 labeled pairs beats the best prompting approach, although it takes more data points to reach a comparable RMSE (fig. 9 in the appendix relates the number of training examples to performance). In addition, finetuning DeBERTa-v3-large on pairwise data results in outputs that strike a balance between correlation and RMSE (fig. 5).

Notably, finetuning a regression model on scalar outputs directly (here, BT estimates induced from the pairwise annotations in our datasets) does not yield the same benefits. Finetuning regression models is less data-efficient and the relation between the number of training examples and RMSE tends to be unstable (see fig. 10).

5 Conclusions

The uptake of LLMs in social science research is high and increasing.¹⁷ For downstream inferences to be valid, it is important to use them appropriately. Our survey on scalar construct measurement aims to guide practitioners, with findings that may translate outside social science constructs.

Limitations

While our datasets are diverse regarding the kind and complexity of social science constructs they cover, they do not span the full breadth of possible scalar constructs. Our study focuses on three social science constructs that represent (political) communication phenomena related to emotions, political attack behaviors, and rhetorical style. However, it is possible that our results and the implications they have for applied researchers may not translate to other conceptually continuous concepts from other

domains, like the level of anxiety expressed in user messages. Moreover, our datasets only include English-language texts with applications focused on U.S. politics. The generalization of our findings to other languages and country contexts is therefore an open question. In such cases, it may be that the gap between prompted models and finetuning is larger, as it is less likely for the constructs to have been attested to in the training data.

Regarding our prompting methods, we note that the results we present focus on few-shot prompting with randomly selected exemplars. We also tested the added value of more strategic exemplar selection, like choosing exemplars at anchoring points of the 1–9 scale. These strategies did not consistently improve the scoring performance in our datasets, however. Moreover, we did not examine the potential added value of Chain-of-Thought (CoT) prompting because Wang et al. (2025) convincingly demonstrated that it can collapse or otherwise disturb the judgment distribution and underlying token probabilities over response options in LLM-as-a-judge applications like ours.

Further, deploying LLMs is compute-intensive, which may hinder practitioners’ adoption of the methods we evaluate. We note, however, that our finding regarding the data efficiency and reliability of small finetuned models paves the way for valid text scoring with smaller, specialized models.

While we cover several open model variants, we do not evaluate closed APIs, which often have a lower barrier to entry for non-technical users. Closed APIs provide no, or only very limited, access to tokens’ generation probabilities, preventing users from using this information for response calibration and debiasing.¹⁸ Further, social science highly values reproducibility, which is at odds with the unpredictable updates and deprecation schedules of closed models (Barrie et al., 2024).

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¹⁷On Scopus (scopus.com), “large language model” yields 1,500 social science articles in 2023 and 5,400 in 2024 (using SUBJAREA ‘SOC1’).

¹⁸Replicating our 5-shot pointwise scoring experiments with OpenAI’s GPT-4o model showed that for typically more than 90% of texts, at least one of the answer candidate tokens’ probabilities was not among the top-20 most likely tokens’ probabilities returned by the API.

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A Methodological details

A.1 The Bradley-Terry model

The Bradley–Terry model turns pairwise comparisons into a single latent scale.

Let the items (e.g., texts) in a set of pairwise comparisons be indexed by $i = 1, \dots, n$. Associate each item i with a positive “strength” parameters $\alpha_i > 0$. In our applications, these parameters estimate items’ locations on the underlying (unobserved) latent dimension that represents the scalar construct under study (e.g., ad negativity).

Then, in a pairwise comparison of i and j , the Bradley-Terry model models the probability that i “beats” j in terms of strength (equivalently, is “higher” on the underlying dimensions) as follows:

$$\Pr(i \succ j) = \frac{\alpha_i}{\alpha_i + \alpha_j}$$

Equivalently, writing $\alpha_i \equiv \exp(\lambda_i)$, with $\lambda_i \in \mathbb{R}$, the log odds of i being chosen over j is the following.

$$\text{logit}(\Pr(i \succ j)) = \log \left[\frac{\Pr(i \succ j)}{\Pr(j \succ i)} \right] = \lambda_i - \lambda_j$$

λ_i represents a latent propensity relevant to the comparison, such as “ability,” “preference,” etc., depending on the context. The outcomes of pairwise comparisons are used to estimate these latent propensities relative to a chosen reference. For identification, one can either choose a reference item or impose a constraint such as $\sum_i \lambda_i = 0$.

After estimation, the parameters may be rescaled (for example, to the unit interval) without affecting the fitted probabilities. Probabilities are typically estimated using maximum likelihood, assuming independence of all pairwise comparisons. Bias-reduced or penalized likelihood methods are often used to handle small samples or quasi- or complete separation.

A.2 Pairwise debiasing

Our pairwise debiasing strategy borrows from Wang et al. (2025) by prompting the model with different orderings of items in pairs to avoid positional biases (Zhao et al., 2021; Han et al., 2023; Wang et al., 2023).

Our pairwise prompts first explain the annotation task that define the criterion for comparison (e.g., emotional intensity) and then present a pair of text items in separate lines prefixed by “Text 1” and “Text 2.” The LLM is tasked to indicate which

of the texts meets the comparison criterion and to respond either with “1” or “2.”

Given this prompt format, our in-context learning approach consists of three steps: augmentation, prompting, and calibration. We first augment our pair-level data by performing two augmentation operations on every pair illustrated in Table 4. In the prompting step, we then present one text pair at a time to the model, tasking it to respond with the label of the text item that is higher on the given comparison dimension (e.g., more emotionally intense).

Next, we generate and record the model’s response, including its token probabilities for the two tokens “1” and “2.” Table 5 for an example of an LLM’s response and token probabilities for the original and augmentation versions of a text pair.

Finally, we debias the model’s response as follows. Let p denote the probability for the token corresponding to label “1” in the model’s generated response. Let i indicate whether the labels were swapped (augmentations 1 and 3), and j whether text items’ order was reversed (augmentations 2 and 3), where $i, j \in \{0, 1\}$ with 0 indicating no swapping/reversing. The model’s preference score for the first text item in a pair is p_{ij} . Specifically, the model’s preference for the first item in a pair is captured by p_{00} , $1 - p_{10}$, p_{01} , and $1 - p_{11}$. That is, given our augmentation strategy, which text item corresponds to choice “1” depends on whether the labels were swapped:

$$y_{ij} = \begin{cases} p_{ij}, & \text{if } i = 0 \text{ (i.e., swapped)} \\ 1 - p_{ij}, & \text{if } i = 1 \text{ (i.e., not swapped)} \end{cases}$$

The *debaised preference score* for the first text item, denoted $\hat{p}_1$, is the average of the *aligned* preferences across the original pair and the three augmented conditions:

$$\hat{p}_1 = \frac{1}{4} \sum_{i=0}^1 \sum_{j=0}^1 y_{ij}$$

The corresponding debaised preference for the second item is $\hat{p}_2 = 1 - \hat{p}_1$. For example, for the pair shown in Table 5, the debaised preference scores are $\hat{p}_1 = 0.688$ and $\hat{p}_2 = 0.312$. The model’s debaised choice is therefore “1”.

			position in prompt	
	swap label	reverse order	first option	second option
<i>original pair</i>	no	no	Text 1: “item 1”	Text 2: “item 2”
augmentation 1	yes	no	Text 2: “item 1”	Text 1: “item 2”
augmentation 2	no	yes	Text 1: “item 2”	Text 2: “item 1”
augmentation 3	yes	yes	Text 2: “item 2”	Text 1: “item 1”

Table 4: Illustration of our pair augmentation strategy.

			Model		
	augmentations		token probs.		choice
	labels swapped	order reversed	“1”	“2”	
<i>original pair</i>	no	no	0.996	0.004	“1”
augmentation 1	yes	no	0.699	0.301	“1”
augmentation 3	no	yes	0.197	0.803	“2”
augmentation 3	yes	yes	0.651	0.349	“1”

Table 5: Illustration of LLM’s preferences for response options and actual response (“choice”) for different augmentations of the same text pair (“original”).

B Experiment details

B.1 GPU specification for LLM inference

We ran LLM inferences on local hardware and the ERH Zurich’s GPU cluster, depending on model size. Table 6 provides details.

Model	Quant.	GPU(s)	GPU RAM
Qwen-2.5-7B	4-bit	1 × NVIDIA GeForce RTX 4090	24,564 MB
Qwen-2.5-32B	4-bit	1 × NVIDIA A100-PCIE-40GB	40,960 MB
	<i>none</i>	3 × NVIDIA GeForce RTX 4090	24,564 MB
Qwen-2.5-72B	4-bit	3 × NVIDIA GeForce RTX 4090	24,564 MB

Table 6: Overview of models, quantization, GPU hardware, and environment.

B.2 Finetuning hyperparameters

We use default hyperparameters for the finetuned models (table 7), except for the learning rate, which we optimize on a validation set. Specifically, we sweep from the order of magnitude below the default to an order of magnitude above, in even steps, maximizing accuracy on a validation set from a distinct pairwise annotation dataset (Benoit et al. 2019, which we do not report on in the main text due to a large amount of annotation noise). The specific values are (**selected in bold**).

- DeBERTa-v3-large: 6.00×10^{-7} , 1.29×10^{-6} , 2.78×10^{-6} , 6.00×10^{-6} , 1.29×10^{-5} , **2.78×10^{-5}** , 6.00×10^{-5}
- Llama-3.1-8B: 2.00×10^{-5} , 4.31×10^{-5} , 9.28×10^{-5} , 2.00×10^{-4} , 4.31×10^{-4} , **9.28×10^{-4}** , 2.00×10^{-3}
- ModernBERT-large: 1.42×10^{-5} , 2.53×10^{-5} , 4.50×10^{-5} , **8.00×10^{-5}** , 1.42×10^{-4} , 2.53×10^{-4} , 4.50×10^{-4}

The maximum sequence length for all models is 384 tokens, which covers over 90% of data points.

Hyperparameter	ModernBERT-large	DeBERTa-v3-large	Llama-3.1-8B
Learning rate	8e-5	2.78e-5	9.28e-4
Epochs	5	10	5
Batch size	32	8	4
Gradient accum. steps	1	8	6
Weight decay	1e-5	–	0.001
Adam β_1	0.9	–	–
Adam β_2	0.98	–	–
Adam ϵ	1e-6	–	–
LoRA r	–	–	8
LoRA α	–	–	32
LoRA dropout	–	–	0.1
Precision	BF16	FP32	BF16
Quantization	–	–	8-bit

Table 7: Finetuning model hyperparameters. – Indicates the default in the Huggingface `transformers` library.

C Additional Results

Figure 6 shows the distribution of LLM confidences in the most-probable (modal) response over multiple model types and prompting strategies. Any given design decision can have a large impact on confidence.

Table 8 compares the pointwise scoring performance of the two open-weights models we use to the performance of GPT-4o based on the Ad-Negativity data. Table 9 shows that our prompting results for pointwise scoring with token probability weighting are robust to running inference without 4-bit quantization.

Complementing the selected results highlighted in Figure 5, Figure 7 shows the relation between model responses and ground-truth BT scores in prompted models’ probability-weighted average pointwise scores (5-shot), and Figure 8 the relation between predicted scores and ground-truth BT scores in models fine-tuned on all examples in datasets’ training splits.

Figure 9, in turn, shows how finetuned models’ classification and scoring performance changes with the number of training examples used for finetuning. Figure 10 contrasts finetuned reward models’ performance with those of regression models finetuned using inferred BT scores instead of pairwise label data.

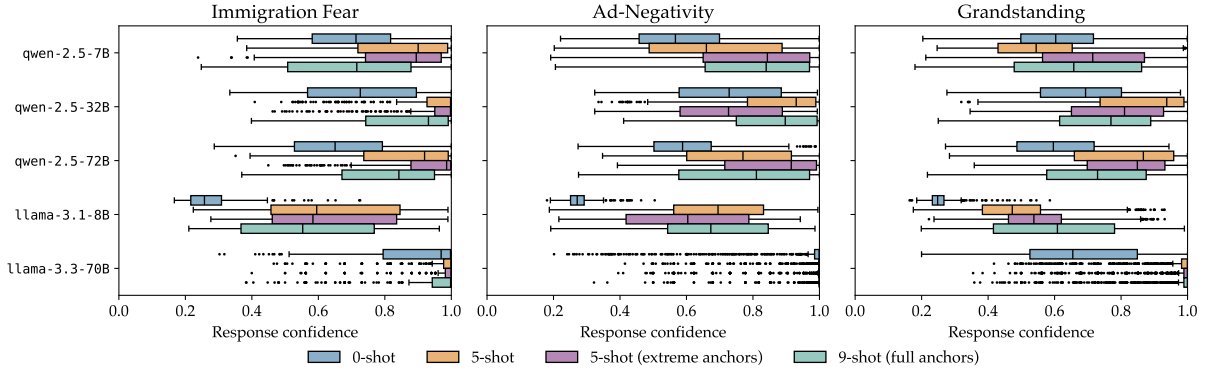


Figure 6: LLM confidence in the modal response varies widely over model variants and prompting strategy, leading to inconsistent behavior that can impact downstream conclusions.

	zero-shot			5-shot		
	Acc	ρ	RMSE	Acc	ρ	RMSE
GPT-4o	0.786 $\pm$ 0.011	0.889 $\pm$ 0.016	0.253 $\pm$ 0.009	0.802 $\pm$ 0.010	0.917 $\pm$ 0.010	0.209 $\pm$ 0.010
Llama-3.3-70b	0.795 $\pm$ 0.011	0.888 $\pm$ 0.014	0.288 $\pm$ 0.008	0.800 $\pm$ 0.010	0.903 $\pm$ 0.017	0.250 $\pm$ 0.010
Qwen-2.5-72b	0.806 $\pm$ 0.010	0.906 $\pm$ 0.011	0.134 $\pm$ 0.007	0.807 $\pm$ 0.012	0.918 $\pm$ 0.010	0.165 $\pm$ 0.011

Table 8: Comparison of pointwise scoring prompting results between open-weights and proprietary models in Ad-Negativity data. *Notes:* Metrics reported are the pair classification accuracy (Acc) and scoring performance relative to ground-truth BT scores in terms of Spearman’s rank correlation (ρ) and the root mean squared error (RMSE, on scale 0–1). Values report are averages $\pm$ one standard deviation computed by summarizing results across five folds.

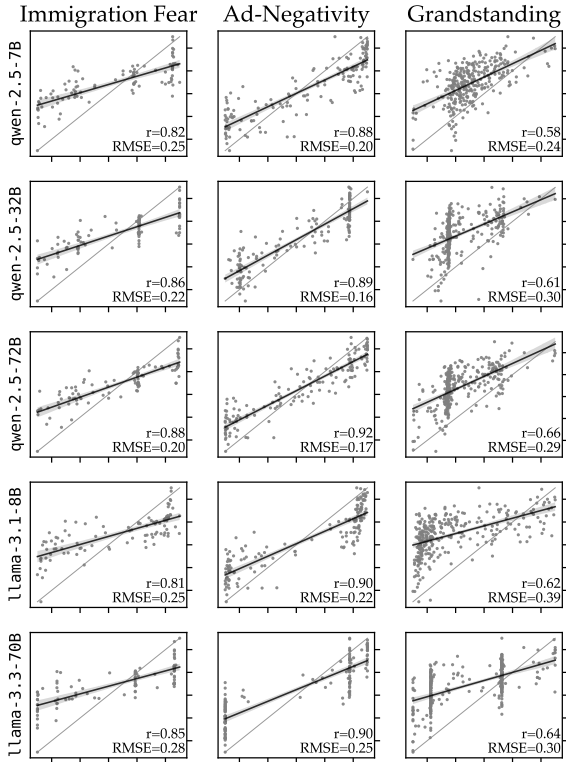


Figure 7: Relation between model responses and ground-truth BT scores in prompted models’ probability-weighted average pointwise scores (5-shot).

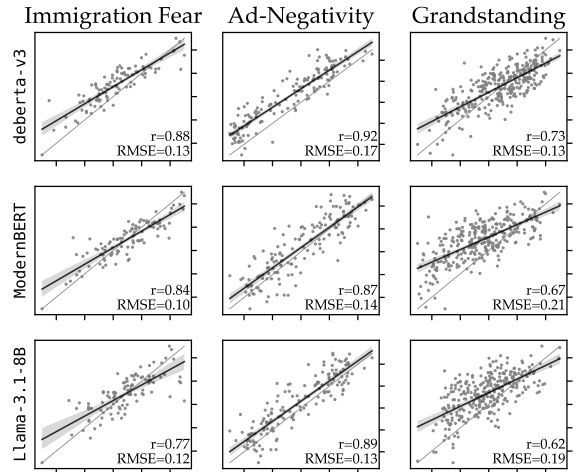


Figure 8: Relation between model responses and ground-truth BT scores in models fine-tuned on all examples in datasets’ training splits (2,729 in the IMMIGRATION FEAR, 6,760 in the AD-NEGATIVITY, and 32,308 in the IMMIGRATION FEARDATA).

	Dataset	Model	Quantization	Acc	ρ	RMSE
IMMIGRATION FEAR	Qwen-2.5-32b		4-bit	0.768 $\pm$ 0.007	0.853 $\pm$ 0.031	0.222 $\pm$ 0.011
			<i>none</i>	0.763 $\pm$ 0.007	0.853 $\pm$ 0.034	0.232 $\pm$ 0.012
	Llama-3.1-8b		4-bit	0.740 $\pm$ 0.008	0.780 $\pm$ 0.037	0.266 $\pm$ 0.011
			<i>none</i>	0.748 $\pm$ 0.008	0.795 $\pm$ 0.034	0.259 $\pm$ 0.012
AD-NEGATIVITY	Qwen-2.5-32b		4-bit	0.799 $\pm$ 0.010	0.896 $\pm$ 0.016	0.158 $\pm$ 0.008
			<i>none</i>	0.789 $\pm$ 0.010	0.897 $\pm$ 0.016	0.163 $\pm$ 0.009
	Llama-3.1-8b		4-bit	0.806 $\pm$ 0.009	0.899 $\pm$ 0.014	0.196 $\pm$ 0.007
			<i>none</i>	0.794 $\pm$ 0.009	0.897 $\pm$ 0.014	0.222 $\pm$ 0.008

Table 9: Comparison of prompting results for 5-shot pointwise scoring with and without 4-bit quantization for selected datasets and models. *Notes:* Rows correspond to the model (e.g., qwen/llama) and few-shot method (0-shot or 5-shot). Metrics reported are the pair classification accuracy (Acc) and scoring performance relative to ground-truth BT scores in terms of Spearman’s rank correlation (ρ) and the root mean squared error (RMSE, on scale 0–1). Values report averages $\pm$ one standard deviation computed based on 25 bootstrapped estimates in test split. Values in bold mark the best result for a dataset and metric and values underlined mark results within one standard error of the best result.

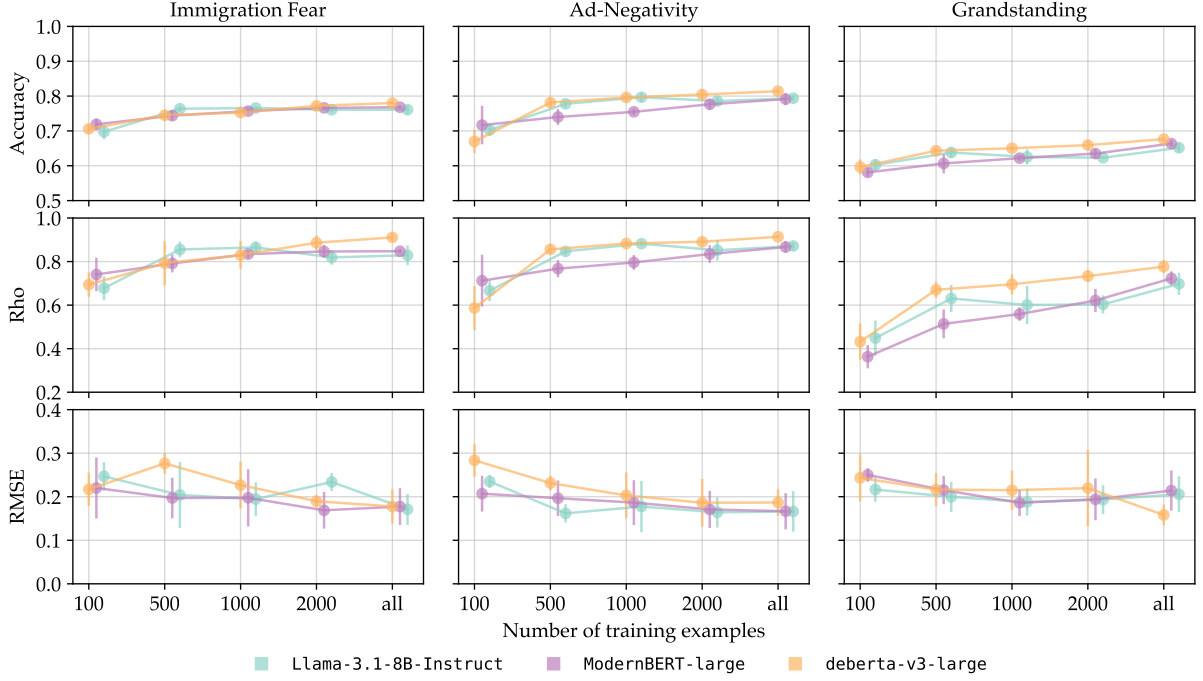


Figure 9: Classification and scoring performance of finetuned models as a function of the number of training examples. “all” refers to all pairs in the training split (2,729 in the IMMIGRATION FEAR, 6,760 in the AD-NEGATIVITY, and 32,308 in the IMMIGRATION FEARDATA). Points and vertical bars report averages $\pm$ one standard deviation computed by summarizing results across five folds.

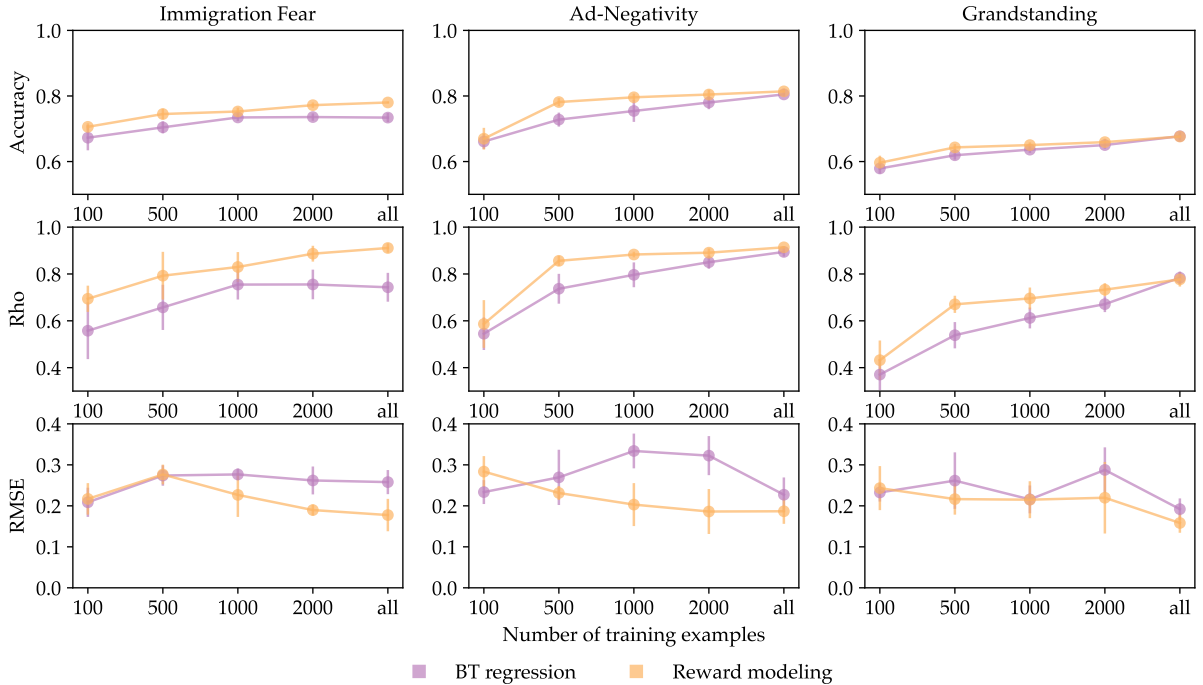


Figure 10: Classification and scoring performance of finetuned reward and regression models as a function of the number of training examples. Results shown for finetunes of deberta-v3-large. “all” refers to all pairs in the training split (2,729 in the IMMIGRATION FEAR, 6,760 in the AD-NEGATIVITY, and 32,308 in the IMMIGRATION FEARDATA). Points and vertical bars report averages $\pm$ one standard deviation computed by summarizing results across five folds.

D Prompts

Pairwise comparison prompt used for IMMIGRATION FEAR task in Carlson and Montgomery (2017) data.

Your task is to **indicate which of these two statements expresses more fear, anxiety, or worry about the negative impact of immigrants or immigration on America.**

Note that we are not interested in whether the writer dislikes immigrants, wants them to go home, resents them, or blames them. We are only interested in whether the writer is expressing fear, anxiety, or worry.

Input

{text}

Response: [Output only "1" or "2"]

Absolute scoring prompt used for IMMIGRATION FEAR task in Carlson and Montgomery (2017) data.

Task

Your task is to **rate the statement below in terms of the level of fear, anxiety, or worry about the negative impact of immigrants or immigration on America expressed in it on a scale from 1 (none) to 9 (extreme).**

Note that we are not interested in whether the writer dislikes immigrants, wants them to go home, resents them, or blames them. We are only interested in whether the writer is expressing fear, anxiety, or worry.

Input

{text}

Response: [Output only a score between 1 and 9]

Pairwise comparison prompt used for AD-NEGATIVITY task in Carlson and Montgomery (2017) data.

Overview

This task involves reading the text of two television advertisements aired during the 2008 U.S. Senate elections. Each advertisement consists of about one paragraph of text. Researchers will use your responses to better understand the "tone" of each political ad.

You will see text from **two advertisements**. Your job is to read both and select the one that is:

- **most** negative towards the candidate(s) mentioned, or;
- **least** positive about the candidate(s) mentioned.

Some of these choices will be very clear, but others will require you to use your best judgement.

Details

Here are a few rules of thumb to guide you:

- Ads that attack a candidate's personal characteristics (e.g., "Bob is dishonest.") are generally more negative than ads that attack a candidate's record or job performance (e.g., "Bob is too liberal.').
- Ads that attack a specific candidate alone (e.g., "Bob is unqualified") are generally more negative than ads that contrast two candidates (e.g., "Bob is unqualified, but Jill is very experienced").
- Ads that attack a named individual (e.g., "Bob spent his time in Washington working for fat cats") are generally more negative than ads that attack a general group (e.g., "We need to stop those fat cats in Washington").
- Ads that state a policy position (e.g., "Bob will find everyone jobs") are generally less positive than ads that praise a candidate as a person (e.g., "Bob is a leader.').
- If both advertisements attack a candidate, pick whichever of the two advertisements is **most** negative.
- If both advertisements praise a candidate, pick whichever of the two advertisements is **least** positive.
- Do not allow your own political opinions to influence your decisions. Your goal is to select the ad that other coders would recognize as the most negative (or least positive).

It is critical that you read each statement carefully. Skimming or reading quickly will result in low-quality evaluations.

Background

The texts you will be reading were collected by the Wisconsin

Advertising Project (WiscAds), which studies political advertisements in the United States. In this task, we are interested in ads from the 2008 U.S. Senate elections.

WiscAds takes each television ad and creates a "storyboard" composed of the words included in the ad's voiceover.

Here are a couple of things to remember:

1. Words in brackets (e.g., [Roberts]) indicate who is speaking. So, when the children speak, it looks like this: '[Kids]: "Right Pat!"'
2. The final line of each ad will always include a bracket [PFB], which is short for "Paid for By." In this case, the ad was paid for by the organization, "Pat Roberts for U.S. Senate." So, in this case, the last line will be: "[PFB:] Pat Roberts for U.S. Senate."
3. The ads you code will not include images. That means you will have to use only the text from the storyboard to code the ads.

Task

Please read the two advertisement texts below. Your job is to read both and select the ad that is:

- **most** `_negative_` towards the candidate(s) mentioned, or;
- **least** `_positive_` about the candidate(s) mentioned.

Input

{text}

Response: [Output only "1" or "2"]

Absolute scoring prompt used for AD-NEGATIVITY task in Carlson and Montgomery (2017) data.

Overview

This task involves reading the text of a television advertisement aired during the 2008 U.S. Senate elections. The advertisement consists of about one paragraph of text. Researchers will use your responses to better understand the "tone" of the political ad.

You will see the text of **an advertisement**. Your job is to read it and rate it on a scale ranging from 1 to 9 in terms of

how:

- 9: `_negative_` it is towards the candidate(s) mentioned, or;
- 1: `_positive_` it is about the candidate(s) mentioned.

Some of these choices will be very clear, but others will require you to use your best judgement.

Details

Here are a few rules of thumb to guide you:

- Ads that attack a candidate's personal characteristics (e.g., "Bob is dishonest.") are generally more negative than ads that attack a candidate's record or job performance (e.g., "Bob is too liberal.').
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2. The final line of each ad will always include a bracket [PFB], which is short for "Paid for By." In this case, the ad was paid for by the organization, "Pat Roberts for U.S. Senate." So, in this case, the last line will be: "[PFB:] Pat Roberts for U.S. Senate."
3. The ads you code will not include images. That means you will have to use only the text from the storyboard to code the ads.

Task

Please read the advertisement text below. Your job is to read it and rate it on a scale ranging from 1 to 9 in terms of how:

- 9: negative it is towards the candidate(s) mentioned, or;
- 1: positive it is about the candidate(s) mentioned.

Input

{text}

****Response**:** [Output only a score between 1 and 9]

Pairwise comparison prompt used for GRANDSTANDING task in Park (2021) data.

Overview

You will be presented two paragraphs from the House representatives' speeches during congressional hearings. Your task is to choose the paragraph that is relatively more opinionized/grandstanding or less factual/information-seeking.

Background

To give you some background knowledge, congressional committees hold hearings for various purposes: to

monitor executive branches, to collect information for legislations, to approve government nominees or budgeting plans, etc. A congressional hearing proceeds as follows: It starts with the committee chair's opening speech followed by other committee members' and witnesses' opening speeches. Then, the chair proceeds to a Q&A session where committee members ask questions to witnesses. Long speeches are broken down to paragraphs. Thus, some paragraphs you will compare can be part of a longer speech.

Details

A speech is an ****opinionized or grandstanding**** speech if it does one of the following:

1. Denouncing (or Praising) a person or an institution (e.g. a party, its members, president, a government agency, a witness or others)
2. Taking positions on a policy by approving or disapproving it (which includes subjective interpretation of a policy-relevant situation)
3. Asking questions just to embarrass or attack a witness:

A speech is a ****factual or information-seeking**** speech if it is one of the following:

1. Objective description of a policy-relevant situation
2. Asking witnesses questions for fact-checking or expert opinion-seeking

A speech is ****neither opinionized nor information-seeking**** if it falls into the following:

1. Procedural remarks:
2. None of these mentioned above (No content):

Important notes

- Consider that speeches can be placed onto a continuum of which one extreme end is opinionized/grandstanding speeches and the other extreme end is factual/information-seeking speeches. In the middle of the two ends, speeches that are neither the two including procedural speeches can be located.
- It is important that you read each

speech extract carefully, and that you judge each by the standards listed above and the information in the text.

- In comparing the two paragraphs, DO NOT make your judgments on your own knowledge of a person or a policy in question or on definitions of opinions different to those listed above.
- Note that not all questions are information-seeking but can be part of grandstanding depending on what is being asked and how. Also, note that the length of a speech excerpt is irrelevant to and does not cue the type of speech.

Task

Please read the two statements below and select the statement that is relatively more opinionized/grandstanding or less factual/information seeking.

- A statement is more opinionized or grandstanding if it denounces or praises an institution or a person, or expresses subjective views on a policy or a situation more explicitly and strongly.
- A statement is factual or information seeking if it gives objective description of a situation or asking witnesses for information or their opinion.

Which of the two statements below is more opinionized/grandstanding or less factual/information seeking?

{text}

****Response**:** [Output only "1" or "2"]

Absolute scoring prompt used for GRANDSTANDING task in Park (2021) data.

Overview

You will be presented a paragraph from the House representatives' speeches during congressional hearings. Your task is rate the paragraph on scale ranging from opinionized/grandstanding on one end to factual/information-seeking on the other.

Background

To give you some background knowledge, congressional committees hold hearings for various purposes: to

monitor executive branches, to collect information for legislations, to approve government nominees or budgeting plans, etc. A congressional hearing proceeds as follows: It starts with the committee chair's opening speech followed by other committee members' and witnesses' opening speeches. Then, the chair proceeds to a Q&A session where committee members ask questions to witnesses. Long speeches are broken down to paragraphs. Thus, the paragraphs you will rate can be part of a longer speech.

Details

A speech is an ****opinionized or grandstanding**** speech if it does one of the following:

1. Denouncing (or Praising) a person or an institution (e.g. a party, its members, president, a government agency, a witness or others)
2. Taking positions on a policy by approving or disapproving it (which includes subjective interpretation of a policy-relevant situation)
3. Asking questions just to embarrass or attack a witness

A speech is a ****factual or information-seeking**** speech if it is one of the following:

1. Objective description of a policy-relevant situation
2. Asking witnesses questions for fact-checking or expert opinion-seeking

A speech is ****neither opinionized nor information-seeking**** if it falls into the following:

1. Procedural remarks:
2. None of these mentioned above (No content):

Important notes

- Consider that speeches can be placed onto a continuum of which one extreme end is opinionized/grandstanding speeches and the other extreme end is factual/information-seeking speeches. In the middle of the two ends, speeches that are neither the two including procedural speeches can be located.
- It is important that you read the

speech extract carefully, and that you judge it by the standards listed above and the information in the text.

- In rating the paragraph, DO NOT make your judgment on your own knowledge of a person or a policy in question or on definitions of opinions different to those listed above.
- Note that not all questions are information-seeking but can be part of grandstanding depending on what is being asked and how. Also, note that the length of a speech excerpt is irrelevant to and does not cue the type of speech.

Task

Please read the statement and rate it on a scale ranging from 1 to 9 how opinionized/grandstanding (9) or factual/information seeking (1).

- A statement is ****opinionized or grandstanding**** if it denounces or praises an institution or a person, or expresses subjective views on a policy or a situation more explicitly and strongly.
- A statement is ****factual or information seeking**** if it gives objective description of a situation or asking witnesses for information or their opinion.

On a scale ranging from 1 to 9, how opinionized/grandstanding (9) or factual/information seeking (1) is the statement shown below?

{text}

****Response****: [Output only a score between 1 and 9]